

Virtual Field Geology Basecamp: Course Summary for Faculty Review

Course Snapshot

This course is a 3-week virtual geology field camp built around digital mapping, structural interpretation, and applied geohazard analysis.

- Format: Fully online or hybrid-compatible
- Length: 3 weeks
- Assignments: 12 graded assignments
- Estimated student workload: about 49.5 hours total
- Total points: 600
- Core platforms:
 - Visible Geology (interactive 2D/3D geology and stereonet work)
 - Leeds Virtual Landscapes (virtual field mapping environments)

Who This Course Is For

This design is a good fit for:

- Introductory to intermediate structural geology and mapping instruction
- Programs needing virtual field alternatives or supplements
- Instructors who want authentic map-plus-cross-section workflows without travel logistics

Students practice professional outputs throughout the course:

- Geological maps with standard cartographic elements
- Cross sections with topographic profiles
- Structural interpretations (including stereonets where appropriate)
- Applied hazard or human-impact reasoning tied to mapped evidence

Learning Design

The course progresses in three stages:

1. Foundations in map-pattern reasoning (contours, dip/strike, Rule of V's)
2. Full virtual landscape mapping (from introductory to advanced structural complexity)
3. Independent capstone project with proposal and final submission

A consistent pattern appears across assignments:

- Observation and model interaction
- Visual evidence submission (screenshots, annotated maps, diagrams)
- Written interpretation using geologic reasoning
- Applied relevance (hazard, infrastructure, or land-use implications)

Assignment Sequence (At a Glance)

Seq	Assignment	Unit	Week	Est. Hours	Points
1	Reading Topographic Maps in 3D	Visible Geology	1	1.5	20
2	Dip Angle and Map Patterns	Visible Geology	1	2.0	25
3	Strike, Downstream Dip, and the Rule of V's	Visible Geology	1	2.5	30
4	Geological Mapping - Lighthouse Bay	Leeds Virtual Landscapes	1	4.0	50
5	Geological Mapping - Three River Hills	Leeds Virtual Landscapes	1	5.0	60
6	Stereonets and Fold Map Patterns	Visible Geology	2	3.0	40
7	Geological Mapping - Rhoscolyn	Leeds Virtual Landscapes	2	7.0	80
8	Basin Stratigraphic Correlation	Visible Geology	2	4.0	60
9	Reservoir Geology and CO2 Sequestration	Visible Geology	2	4.5	55
10	Building a 3D Geological Model	Visible Geology	3	4.0	50
11	Independent Mapping Project - Proposal	CYOA Final Project	3	2.0	30
12	Independent Mapping Project - Final Submission	CYOA Final Project	3	10.0	100

Weekly Arc and Workload

Week 1 (Foundations and Introductory Mapping)

- Hours: 15.0
- Points: 185
- Focus:
 - Topographic interpretation in 2D/3D
 - Dip/strike effects on outcrop pattern
 - Rule of V's mastery
 - Intro/intermediate map and cross-section production in virtual landscapes

Week 2 (Advanced Structural and Basin Analysis)

- Hours: 18.5
- Points: 235
- Focus:
 - Stereonet interpretation and fold geometry
 - Complex real-world mapping (Rhoscolyn)
 - Basin-scale stratigraphic reasoning and drill-core correlation
 - Reservoir/seal logic and CO2 storage viability analysis

Week 3 (Synthesis and Independent Work)

- Hours: 16.0
- Points: 180
- Focus:
 - Student-built 3D model integrating map and section logic
 - Capstone proposal and execution
 - Final integrated deliverable set with hazard or human-impact assessment

Assessment Model

Assessment is rubric-driven and artifact-rich. Most assignments grade a combination of:

- Technical accuracy (map boundaries, symbols, structural consistency)
- Subsurface reasoning quality (cross sections and geometry)
- Structural interpretation (including stereonet products when required)
- Application quality (hazard/impact assessment with specific map evidence)
- Communication and presentation standards

The capstone final project is the largest single assessment (100 points) and requires integrated geological reasoning across all major skill areas.

Capstone Flexibility

The final project supports multiple pathways:

- Remote structural geology
- Remote geomorphic or Quaternary mapping
- Remote lithologic mapping
- Field structural geology (optional, on public land)
- Field geomorphic or engineering geology

All pathways require at least:

- Geological map
- Geological cross section
- Human-impact or geohazard assessment
- Stereonet, when the selected project type is structural

Adoption Considerations for Another Professor

Strengths

- Strong scaffold from fundamentals to independent synthesis
- Authentic outputs aligned with field-geology practice
- Built-in hazard/impact framing improves real-world relevance
- Flexible capstone supports different student interests and contexts

Requirements

- Instructor familiarity with basic structural geology mapping conventions
- Access to Visible Geology and Leeds Virtual Landscape resources
- Student support for screenshot/PDF submission workflow
- Time for rubric-based review of maps, sections, and written interpretation

Typical Implementation Pattern

A professor can adopt this as:

- A compact 3-week intensive
- A module sequence embedded within a longer structural geology course
- A virtual alternative when field access is constrained

Bottom Line

This is a high-structure, evidence-driven virtual field course that emphasizes practical mapping competence, structural interpretation, and applied geologic decision-making. It is especially attractive for instructors who want rigorous field-style outcomes in an online or hybrid format without sacrificing authentic geoscience workflows.